The Relational Nature of Information

Why Information Is Neither Matter nor Energy

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Abstract

Information is commonly treated either as a measurable quantity (Shannon), a semantic entity (Floridi), or a physical resource ("It from Bit"). This paper proposes a different ontological perspective: information is not an intrinsic property of matter, nor of structure alone, but an emergent property of a relation established between systems through a shared structural code.

The framework separates four conceptual layers — matter, structure, coding, and information — and further distinguishes two kinds of coding relation: **coding-by-convention**, which requires agents and historical agreement, and **coding-by-mechanism**, which requires only causal machinery. This distinction explains why identical physical configurations may carry information for one system while remaining meaningless for another, and clarifies how the present proposal differs from — rather than restates — Bateson's notion of "a difference that makes a difference" and Floridi's semantic theory of information.

1. Introduction

What is information?

Despite decades of research, no universally accepted ontological definition exists.

Three major traditions dominate contemporary thought.

- Shannon identifies information with statistical uncertainty.
- Physics increasingly treats information as a conserved physical quantity.
- Philosophy often defines information through meaning or semantic content.

Each perspective captures an important aspect of information. Yet each leaves a question unresolved:

> *Where does information actually exist?*

This paper argues that information does not exist as an intrinsic property of objects. Instead, it emerges from structural relations between systems.

Two relational accounts already exist in the literature: Bateson's characterization of information as "a difference that makes a difference," and Floridi's semantic theory, which grounds information in well-formed, meaningful, and truthful data relative to an interpreting system. Before introducing a new formalism, it is fair to ask whether this paper restates either view in different vocabulary. Section 8 answers this question directly. The short version: this paper's relation $R(S, C)$ is narrower than Bateson's (it requires a specifiable code, not just any difference) and weaker than Floridi's (it does not require truth or semantic content). The four-layer decomposition exists to make that comparison precise rather than rhetorical.

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2. The Problem

Consider a modem transmitting a signal.

The wire contains electrons. The signal possesses amplitude, frequency, and phase. Physics completely describes these quantities.

But nowhere inside Maxwell's equations does the binary sequence appear:

$$101101$$

The bits appear only because transmitter and receiver share a coding protocol.

Without that protocol, the same electromagnetic wave becomes merely a physical oscillation.

The physical structure remains identical. The information disappears.

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3. Four Ontological Layers

The confusion surrounding information arises because four different categories are often conflated.

Layer 1 — Matter

Matter provides the physical substrate. Examples: electrons, photons, molecules, neurons. Matter exists independently of interpretation.

Matter alone, regardless of quantity, does not constitute order. A nebula contains more matter than a crystal; it is not, for that reason, more structured. Layer 1 sets no constraint on Layer 2 — quantity of matter is orthogonal to degree of structure. Without Structure, matter is chaos, however much of it there is.

Layer 2 — Structure

Structure organizes matter. Examples: waveforms, molecular geometry, neural connectivity, DNA sequences. Structure also exists objectively.

Layer 3 — Coding

Coding establishes correspondences between structures. The first version of this paper treated coding as a single, homogeneous category. That conflation is itself a source of confusion, because the examples invoked — a modem protocol and the genetic code — are not the same kind of correspondence. The revised framework distinguishes two subtypes.

3.1 Coding-by-convention

A correspondence established and sustained by agreement among agents capable of revising it. Examples: a modem protocol (phase 180° → binary 1), a natural language, musical notation. These codes are historically contingent and require something on the agent side — intention, convention, the possibility of renegotiation.

3.2 Coding-by-mechanism

A correspondence sustained by causal machinery, with no agents and no possibility of renegotiation by the systems involved. Example: the genetic code (codon AUG → methionine), implemented by ribosomes and aminoacyl-tRNA synthetases. Nothing in this correspondence requires intention on the part of the cell.

Both subtypes are coding relations in the sense relevant to this paper: both are relational mappings, neither reducible to matter nor to structure alone. But collapsing them into one category obscures a question that the genetics and philosophy-of-biology literature has long debated — whether talk of genetic "code" and "information" is anything more than a useful metaphor for differential causal specificity. This paper takes no side in that debate. What it claims is narrower: if coding-by-mechanism is genuine coding, the relational account below applies to it without modification, because the account never required agents in the first place. This is a strength of the relational view, not a complication: it explains DNA and modems with the same minimal apparatus, without needing to import semantic notions like meaning or belief into ribosomal chemistry.

3.3 Latent vs. accessible coding

$R(S, C)$ as stated in Sections 3.1–3.2 conflates two distinct claims: that a coding relation C *exists*, and that C is *available now* to some interpreting system. These can come apart, and ordinary cases force the distinction.

Consider a text written in a script whose code is no longer held by any living reader. The marks on the page are Structure (Layer 2) — that much is uncontroversial. The question is whether they constitute Information (Layer 4) in the absence of any system currently able to apply C. Two answers are available, and they conflict: either the text has *always* carried information, because C was fixed by its writers regardless of who can use it today; or the text carries no information *until* C is reconstructed, because $R(S, C)$ requires the relation to actually be in force. Most people's intuitions side with the first answer — and $R(S, C)$ as originally stated cannot express why, since it has no place for a code that exists but is not presently exercised.

The fix is to index C temporally rather than treat it as a single fixed relation:

$$\text{Information} = R(S, C_t)$$

where C_t ranges over three states:

STATE OF C_t	DESCRIPTION	IS THERE INFORMATION?
Active	C is held and exercised by some system now	Yes, ontologically and epistemically
Latent	C was fixed at creation but is not currently held by any interpreting system; it is recoverable in principle (by decipherment, comparative reconstruction, or surviving partial transmission)	Yes, ontologically — no, epistemically, until recovered
Indeterminate or lost	No system is known ever to have held C, or recovery is not possible even in principle	Unresolved or no

This is not a re-definition of the Relational Information Principle stated in Section 4 — it is the same principle made precise about *when* C must hold. $R(S, C)$ remains the operative relation; C_t only specifies that C need not be exercised at the moment of inquiry to have been fixed.

Worked case: the Chilam Balam books. The colonial-era Maya Chilam Balam texts are written in Yucatec Maya using the Latin alphabet, not in the pre-Columbian hieroglyphic script. The code — Yucatec Maya grammar and vocabulary, continuously transmitted through a living language community — was never fully lost; specialists read these texts today, even where individual passages remain contested. This is the **latent, largely recovered** case: C_t was briefly attenuated (colonial disruption, archaic vocabulary, allusive style) but never fully severed from a living tradition. The texts had information throughout; access to that information varied over time. Contrast this with the pre-Columbian hieroglyphic corpus itself, large portions of which became genuinely **latent and unrecovered** until twentieth-century decipherment work — a clearer case of C_t moving from inaccessible to active without S ever changing.

Worked case: undeciphered scripts. Linear A and the Rongorongo tablets are sharper instances of the latent state with recovery still unresolved. Structure is not in dispute — both show patterning consistent with writing (constrained symbol inventories, repetition, statistical regularities atypical of decoration). What is unresolved is whether C is recoverable even in principle, given the small corpus size and absence of a bilingual key comparable to the Rosetta Stone. The honest answer the framework gives is: *latent, with recovery status open* — not "no information," and not "confirmed information," but a third category the original two-state version of $R(S, C)$ had no room for.

Worked case: the Wow! signal. This case differs from the two above in a way worth marking explicitly: with Chilam Balam and Linear A, C's prior existence is not in question — humans wrote these texts, so a code was fixed at creation even where recovery is incomplete. For the Wow! signal, received in 1977 as a 72-second narrowband burst from the direction of Sagittarius and never repeated, *whether C ever existed at all* is precisely what is unknown. The structural anomaly — band narrowness atypical of known natural sources — is evidence that S is *consistent with* an artificial origin, but consistency with artificiality is not evidence that a C was ever fixed by anything. This is the **indeterminate** state, not the latent one: not "a code we lost," but "a structure we cannot yet classify as coded or uncoded." The same indeterminacy applies to the open question of Section 7 — neural structure and experience are in the identical position, an S for which no C has been confirmed, with the further question of whether the search is for a *latent* code (not yet found, like Linear A) or whether the case is *outside Layer 4 entirely* (no C to find, structurally analogous to ruling out an artificial origin for the Wow! signal).

The convention/mechanism split (3.1–3.2) and the active/latent/indeterminate split (3.3) are independent axes. A code can be conventional and latent (Linear A, presumably a human convention now unrecoverable), mechanistic and active (the genetic code, continuously exercised), or — the open cases — of unknown type altogether, because classifying C as conventional or mechanistic already presupposes that C exists, which is exactly what is unresolved for the Wow! signal and for consciousness.

Layer 4 — Information

Information appears only when a structure is interpreted through a shared code — of either subtype. Therefore:

$$\text{Information} \neq \text{Matter}$$

$$\text{Information} \neq \text{Structure}$$

$$\text{Information} = \text{Structure} + \text{Code} + \text{Relation}$$

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4. The Relational Information Principle

We propose:

> ***Relational Information Principle.*** *Information is not an intrinsic property of physical systems. It is an emergent property of relations established through shared structural codes — codes that may be conventional or mechanistic.*

This can be written compactly as notation, not as a derivation:

$$\text{Information} = R(S, C)$$

where S is a physical structure and C is a coding relation (by convention or by mechanism). Without C, only structure exists.

A note on what this notation does and does not do. $R(S, C)$ is shorthand for the claim already stated in prose: information requires both a structure and a code. It is not a derivation engine. No further theorem is extracted from the symbol R that was not already asserted in the Relational Information Principle itself. The notation earns its place only by making cross-domain comparison easier (Section 5) and by giving Section 6 a fixed vocabulary to negate. Readers should not expect $R(S, C)$ to predict, measure, or quantify information — that would require specifying R's internal structure (composability, a metric, a relationship to Shannon entropy), which is left as open work in Section 9.

Section 3.3 refines this further: C need not be exercised at the moment of inquiry for information to exist. $R(S, C_t)$, with C_t ranging over active, latent, and indeterminate states, is the operative form of the principle used from here on; $R(S, C)$ is retained in the text below wherever the temporal state is not at issue.

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5. Examples

Each example is labeled by coding subtype, following Section 3.

Telecommunications — coding-by-convention

A modem signal carries information only because transmitter and receiver negotiated the same protocol. Without synchronization, only electromagnetic waves exist.

DNA — coding-by-mechanism

DNA contains nucleotides. Nucleotides become genetic information only because cellular machinery implements the genetic code. Without ribosomes, DNA becomes chemistry. No agreement between molecules is required — only the causal apparatus that fixes the correspondence.

Human language — coding-by-convention

A Chinese book contains ink patterns. For someone unable to read Chinese, those patterns remain physical structures. Information appears only for readers sharing the linguistic code — a code maintained by a community of speakers who could, in principle, change it.

Music — coding-by-convention

A musical score is paper and ink. Only musicians sharing musical notation recover the music.

Computing — coding-by-convention, mechanically enforced

Machine code consists of voltage levels. Processors interpret those voltages according to an instruction set architecture. The architecture itself originates as a human convention, but once fixed in hardware it is enforced mechanically, with no renegotiation at the level of the running system. This is a hybrid case: conventional in origin, mechanistic in operation. Without the architecture, there is no software.

6. Consequences

The relational view explains why identical physical structures may simultaneously contain information and contain none, depending on whether a coding relation — of either subtype — is in force. Information therefore cannot be reduced either to matter or to geometry alone.

The convention/mechanism distinction adds a further consequence: the relational account does not depend on the presence of intentional agents. This separates it from theories of information that build in a requirement of meaning, belief, or truth from the start (see Section 8).

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7. Consciousness: Where the Coding Relation Is Unknown

The same apparatus can be used to locate — not dissolve — a known difficulty. Neural activity constitutes physical structure (Layer 2 in Section 3). The hard problem of consciousness can be restated within this framework as follows: in every example of Section 5, the coding relation C is identifiable — a negotiated protocol, a ribosome, a linguistic community, a notational convention. For consciousness, no comparable C has been identified that would relate neural structure to experience as $\text{Information} = R(S, C)$ relates a waveform to a bit.

This does not explain experience. It relocates the open question: is there a coding relation linking neural structure to experience that has simply not yet been found, in which case the hard problem is continuous with the other cases in this paper — or is experience not a case of Layer 4 at all, in which case the relational framework has nothing to say about it, and the hard problem lies outside its scope. The present paper does not adjudicate between these two options. What it can say is only this: a complete neural map is a description of Structure. Whether that description stands in any C-relation to experience is exactly the question the relational framework leaves open, rather than one it answers.

This is structurally the same indeterminacy discussed for the Wow! signal in Section 3.3: an S that is anomalous or rich enough to invite the hypothesis of a code, without a confirmed C. Both cases sit in the indeterminate row of the table in 3.3, not the latent row — the open question is not "where did we put the key," as with Linear A, but "is there a lock here at all."

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8. Discussion

This proposal differs from existing approaches in ways worth stating precisely rather than by general contrast alone.

Shannon, physics, and semantic theories

Shannon measures uncertainty. Physics studies state evolution. Semantic theories analyze meaning. The present proposal addresses a different question: the ontology of information itself — not how much there is, not how physical states evolve, but what information is, prior to measurement or meaning.

Bateson

Bateson's "a difference that makes a difference" is already a relational account, and it predates the present proposal by decades. The relation to it should be stated directly rather than left implicit. Bateson's formula does not distinguish the structure that carries a difference from the code that makes that difference count as a difference for some receiving system; "making a difference" folds both into one relation. The present framework separates them: Structure (Layer 2) is what could carry a difference, Coding (Layer 3) is what fixes whether and how it counts as one, and Information (Layer 4) is the resulting relation. This separation has a payoff Bateson's formula does not: it can express the modem case at the moment synchronization is lost. The waveform still carries a difference in the physics sense — it is not constant — but with no protocol in force, that difference makes no difference to any receiver, which the present framework expresses as $R(S, \varnothing)$ and Bateson's formula must instead treat as simply no information, without isolating why.

Floridi

Floridi's semantic theory of information requires data to be well-formed, meaningful, and (for factual information specifically) truthful, relative to a system capable of holding informational states. This is a stronger requirement than the one proposed here. Coding-by-mechanism, as in the DNA case, involves no truth-bearing states and no system that could be said to believe anything; a ribosome does not represent methionine as true. If Floridi's truth condition is taken as necessary for information in general, the DNA case either is not information at all, or requires a separate, weaker notion to cover it. The present framework's $R(S, C)$ is deliberately weaker than Floridi's semantic information: it requires only a fixed correspondence, not truth or belief. The cost of this weaker requirement is that $R(S, C)$ by itself does not distinguish true from false, or meaningful from nonsensical, within an established code — it only marks the boundary between codeless structure and coded structure. That distinction is exactly what is needed for the modem and DNA cases, and exactly what is missing for them in Floridi's framework, which was built primarily with semantic, agent-relative cases in mind.

Summary of the comparison

	REQUIRES AGENTS	REQUIRES TRUTH/MEANING	COVERS DNA WITHOUT METAPHOR
Bateson (difference)	No	No	Partially — does not separate structure from code
Floridi (semantic info.)	Yes	Yes	No — requires reinterpretation
This paper, $R(S, C)$	No	No	Yes — by design

Information is neither an object nor a substance. It is a relational property emerging whenever structural configurations become mutually interpretable through a specifiable code — whether that code is held by agents or implemented mechanically.

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9. Conclusion

Matter carries physical states. Structure organizes matter. Coding — conventional or mechanistic — establishes correspondences. Information emerges only through the relation between a structure and a code.

Information is not "stored" inside matter. Matter supports structures; structures enable coding; coding allows information to emerge. Information is best understood not as a thing, but as a relation — one whose two subtypes (convention and mechanism) share a common form, $R(S, C)$, without sharing a common origin.

Three items are left explicitly open rather than papered over. First, $R(S, C)$ as presented is notation, not a quantitative theory; recovering Shannon entropy as a special case of R — where C is held fixed and S varies stochastically — is a natural next step, not yet carried out here. Second, whether consciousness falls under Layer 4 at all (Section 7) is left unresolved. Third, Section 3.3's indeterminate state has no decision procedure attached: the framework distinguishes latent codes (Linear A) from indeterminate structures (the Wow! signal, consciousness) by description, but does not say how a given case should be classified, or how confidence should move a case from indeterminate toward either active or ruled-out. All three are flagged as the natural continuation of this paper rather than absorbed into premature claims of generality.

Appendix A — Correspondence with TNA

The relational framework presented in this paper is intentionally independent of the formal machinery developed in the **Theory of Axiomatic Necessity (TNA)**. The correspondence below is offered as a structural mapping between two complementary descriptive levels, not as an identity between concepts. Each row is justified individually rather than asserted by table alone, since the original draft's row-by-row correspondence (in particular, Coding ↔ Coherence Restriction) was not argued for and admits at least one equally plausible alternative.

RELATIONAL INFORMATION FRAMEWORK	TNA CORRESPONDENCE	JUSTIFICATION
Matter	Physical substrate (pre-formal domain)	Both denote what exists prior to any formal or structural description.
Structure	$P(\Omega)$: projected structural configurations	Both denote admissible organization independent of whether it is ever realized or interpreted.
Coding	R: Coherence Restriction	Both filter which structures are eligible to be realized/interpreted — see caveat below.
Interpretation	I: Instantiation Operator	Both denote the act that converts an eligible structure into something realized within a domain.
Information	Relational realization within N_1	Both denote the product that exists only after the preceding stages have all occurred.

Caveat on the Coding ↔ R correspondence. An equally defensible mapping would assign Coding to the Instantiation Operator I instead: I is the operator that performs realization, which resembles what a code does (turning structure into something interpreted), whereas the Coherence Restriction R filters admissibility prior to interpretation — arguably closer to a constraint on Structure than to Coding itself. This paper adopts Coding ↔ R because both, in their respective frameworks, are described as conditions for eligibility that must hold before any instantiation/interpretation occurs; but the alternative mapping is not refuted, only set aside as a modeling choice. Readers should treat the table as one consistent mapping among more than one available, not as a derived result.

Compatibility with TNA's Principle of Non-Derivability

Projected structures may exist independently of realization, but realization cannot be recovered from structural description alone. Likewise, physical structures may exist independently of information, but information cannot emerge without the relational coding required for interpretation. Both theories distinguish structural existence from relational realization, while addressing different aspects of that distinction.

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